

Olympiad Test Paper — Science (Class 7)

Chapter 13: Motion and Time

Subject	Science (NCERT Class 7)
Chapter	13 — Motion and Time
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Use the conversions $1 \text{ km/h} = 5/18 \text{ m/s}$ and $1 \text{ m/s} = 3.6 \text{ km/h}$. No calculator is needed — all numbers are designed to work by hand. Most questions need two to four reasoning steps; watch for traps. Do not write on this paper — mark answers on your answer sheet.

1. Which of the following is an example of **periodic** motion that is **not** oscillatory?

- (A) A swinging pendulum (B) A vibrating guitar string
(C) The Earth revolving around the Sun (D) A child going to-and-fro on a swing

2. A bus covers the first 100 m in 5 s and the next 100 m in 4 s. Its motion is:

- (A) uniform, because the distances are equal
(B) non-uniform, because equal distances take unequal times
(C) uniform, because it moves in a straight line (D) at rest throughout

3. A snail crawls 6 m in 2 minutes. Its speed is:

- (A) 0.05 m/s (B) 0.5 m/s
(C) 3 m/s (D) 0.1 m/s

4. Convert 72 km/h into m/s.

- (A) 25 m/s (B) 200 m/s
(C) 18 m/s (D) 20 m/s

5. Convert 15 m/s into km/h.

- (A) 45 km/h (B) 54 km/h
(C) 540 km/h (D) 4.17 km/h

6. A car goes from town A to town B at 60 km/h and returns along the same road at 40 km/h. Its **average speed for the whole trip** is:

(A) 50 km/h

(B) 100 km/h

(C) 24 km/h

(D) 48 km/h

7. Anya and Ravi both leave home for a town 60 km away. Anya drives at a steady 40 km/h the whole way. Ravi drives the first 30 km at 30 km/h and the last 30 km at 60 km/h. Who reaches the town first?

(A) Anya arrives first

(B) Ravi arrives first

(C) they arrive at the same time

(D) it cannot be determined

8. A cyclist's distance–time graph has three parts: 0–10 s a straight slanted line, 10–20 s a horizontal line, 20–30 s a steeper slanted line. The cyclist:

(A) moves, then stops, then moves faster (B) speeds up the whole time

(C) moves, then slows down, then stops (D) stays at rest throughout

9. A pendulum completes 20 oscillations in 40 s. Its time period is:

(A) 2 s

(B) 0.5 s

(C) 20 s

(D) 800 s

10. The SI unit of time is the:

(A) minute

(B) hour

(C) second

(D) day

11. A train moves at 90 km/h. How far does it travel in 10 seconds?

(A) 900 m

(B) 150 m

(C) 25 m

(D) 250 m

12. A cyclist rides 30 km in 2 h, rests for 1 h, then rides 20 km in 1 h. The average speed for the whole journey is:

(A) 16.7 km/h

(B) 12.5 km/h

(C) 25 km/h

(D) 10 km/h

13. Increasing the length of a simple pendulum will:

(A) decrease its time period

(B) leave its time period unchanged

(C) stop it from swinging

(D) increase its time period

14. Which ancient device tells time using the position of the Sun's shadow?

(A) sundial

(B) water clock

(C) sand clock

(D) pendulum clock

15. Two objects P and Q are plotted as straight lines from the origin on a distance–time graph. P is steeper than Q. Which object is faster?

(A) Q

(B) P

(C) both equal

(D) cannot be decided

16. Object X moves at 36 km/h; object Y moves at 12 m/s. Which is faster?

- (A) X (B) both equal
(C) Y (D) cannot compare different units

17. A car travels at 54 km/h for 20 minutes. The distance covered is:

- (A) 27 km (B) 1080 km
(C) 18 km (D) 9 km

18. How long does a 240 km journey take at a steady 80 km/h?

- (A) 3 h (B) 320 h
(C) 2 h (D) 0.33 h

19. Which of these timekeepers would **fail** to work on a cloudy night?

- (A) water clock (B) sand clock
(C) pendulum clock (D) sundial

20. A stone tied to a string and whirled around in a circle shows:

- (A) rectilinear motion (B) circular motion
(C) oscillatory motion (D) random motion

21. A distance–time line that curves upward (its slope keeps increasing) indicates:

- (A) uniform motion (B) non-uniform motion with increasing speed
(C) rest (D) backward motion

22. A car travels at 40 km/h for 2 h and then at 60 km/h for 2 h. Its average speed for the whole trip is:

- (A) 48 km/h (B) 100 km/h
(C) 50 km/h (D) 52 km/h

23. Two pendulums have the **same length** but **different bob masses**. Their time periods are:

- (A) equal (B) larger for the heavier bob
(C) larger for the lighter bob (D) zero for the heavier bob

24. A table shows time (s): 0, 1, 2, 3, 4 and distance (m): 0, 5, 10, 15, 20. The motion is:

- (A) 20 m/s, uniform (B) 5 m/s, non-uniform
(C) speeding up (D) 5 m/s, uniform

25. A table shows time (s): 0, 1, 2, 3, 4 and distance (m): 0, 10, 20, 20, 20. Between $t = 2$ s and $t = 4$ s the object is:

- (A) moving at 10 m/s (B) moving backward
(C) speeding up (D) at rest

26. A pendulum makes 30 oscillations in 60 s. Its string is then lengthened so that its time period becomes 3 s. How many oscillations will it now make in 60 s?

- (A) 30 (B) 20
(C) 45 (D) 90

27. Three runners move at these speeds: P at 5 m/s, Q at 21.6 km/h, R covers 300 m in 1 minute. Who is the fastest?

- (A) P (B) R
(C) Q (D) all three are equal

28. A car covers 360 km: the first 180 km at 90 km/h and the next 180 km at 60 km/h. Its average speed is:

- (A) 72 km/h (B) 75 km/h
(C) 70 km/h (D) 80 km/h

29. Arrange these three speeds in increasing order: 10 m/s, 30 km/h, 720 m/min.

- (A) 30 km/h < 10 m/s < 720 m/min (B) 10 m/s < 30 km/h < 720 m/min
(C) 720 m/min < 10 m/s < 30 km/h (D) all three are equal

30. Pendulum X is longer than pendulum Y. In one minute, which one makes **more** oscillations?

- (A) X (B) both equal
(C) it depends on the bob mass (D) Y

31. A sprinter runs 100 m in 12.5 s. The average speed is:

- (A) 12.5 m/s (B) 8 m/s
(C) 80 m/s (D) 0.125 m/s

32. A car moves at a steady 54 km/h. How many **metres** does it cover in 2 minutes?

- (A) 108 m (B) 6480 m
(C) 1800 m (D) 900 m

33. A ball rolling **up** a slope covers 5 m in the 1st second, 3 m in the 2nd second and 1 m in the 3rd second. Its motion is:

- (A) uniform (B) non-uniform, speeding up
(C) non-uniform, slowing down (D) at rest

34. Two cars start from the same point. On a distance–time graph their lines cross at $t = 3$ s. At that instant the two cars:

- (A) have the same speed (B) both stop
(C) reverse direction (D) are at the same distance from the start

35. Two cars cover the same distance: car A takes 2 h and car B takes 3 h. Which is faster, and in what ratio are their speeds (A : B)?

- (A) A is faster, 3 : 2 (B) B is faster, 2 : 3
(C) A is faster, 2 : 3 (D) they are equally fast

36. An odometer fitted in a vehicle measures:

- (A) speed
- (C) time

- (B) total distance travelled
- (D) fuel level

37. On a distance–time graph, a curve that becomes **less steep** (flattens out) over time means the object is:

- (A) speeding up
- (B) slowing down
- (C) at rest
- (D) moving at constant speed

38. A train runs 120 km in 2 h, halts 20 minutes at a station, then runs another 90 km in 1 h 40 min. Its average speed for the whole trip is:

- (A) 52.5 km/h
- (B) 60 km/h
- (C) 70 km/h
- (D) 63 km/h

39. Which statement about motion types is correct?

- (A) every periodic motion is also oscillatory
- (B) "periodic" and "oscillatory" mean exactly the same thing
- (C) no oscillatory motion is periodic
- (D) every oscillatory motion is periodic, but not every periodic motion is oscillatory

40. A distance–time graph line passes through (0 s, 0 m) and (4 s, 20 m). The speed is:

- (A) 4 m/s
- (B) 20 m/s
- (C) 5 m/s
- (D) 80 m/s

41. A distance–time graph line passes through (2 s, 6 m) and (8 s, 30 m). The speed is:

- (A) 5 m/s
- (B) 6 m/s
- (C) 3 m/s
- (D) 4 m/s

42. A cheetah runs at a steady 30 m/s. How long does it take to cover 540 m, and what is this speed in km/h?

- (A) 18 s and 30 km/h
- (B) 9 s and 108 km/h
- (C) 18 s and 108 km/h
- (D) 18 s and 8.3 km/h

43. A pendulum makes 30 complete oscillations in 45 s. How many complete oscillations will it make in 3 minutes?

- (A) 90
- (B) 120
- (C) 60
- (D) 200

44. A car travels 60 km in 1 h, stops for 30 min, then travels another 60 km in 1 h. Its average speed for the whole trip is:

- (A) 48 km/h
- (B) 60 km/h
- (C) 40 km/h
- (D) 50 km/h

45. A train moving on a long straight track exhibits:

- (A) rectilinear motion
- (B) circular motion
- (C) oscillatory motion
- (D) random motion

46. To find the **average speed** of a car over a whole trip, the readings you need come from:

- (A) the speedometer only
- (B) the odometer and a clock
- (C) a clock only
- (D) a sundial

47. A table shows time (s): 0, 2, 4, 6 and distance (m): 0, 8, 16, 24. The motion is:

- (A) non-uniform, 4 m/s
- (B) uniform, 8 m/s
- (C) uniform, 4 m/s
- (D) uniform, 2 m/s

48. Car P covers 60 km in 1.5 h; car Q covers 80 km in 2 h. Then:

- (A) P is faster
- (B) Q is faster
- (C) it cannot be decided
- (D) both have equal speed

49. A car at 90 km/h and a bike at 22 m/s travel toward each other. Which is faster, and by how much?

- (A) the bike, by 3 m/s
- (B) they are equally fast
- (C) the car, by 3 m/s
- (D) the car, by 68 m/s

50. A table shows time (s): 0, 1, 2, 3 and distance (m): 0, 2, 8, 18. The motion is:

- (A) non-uniform, speeding up
- (B) uniform, 6 m/s
- (C) uniform, 2 m/s
- (D) at rest

51. A ceiling-fan blade (switched on) and a swinging pendulum show, respectively:

- (A) oscillatory motion and circular motion
- (B) circular motion and oscillatory motion
- (C) both circular motion
- (D) both oscillatory motion

52. On a distance–time graph (distance measured from the start), a line that rises and then comes back down to the time axis means the object:

- (A) sped up continuously
- (B) stopped permanently
- (C) moved in a circle
- (D) returned to its starting point

53. A swimmer covers 1.5 km in 25 minutes. Her average speed in m/s is:

- (A) 60 m/s
- (B) 1.0 m/s
- (C) 0.06 m/s
- (D) 3.6 m/s

54. Light travels about 300000 km in 1 s. The distance it covers in 5 s is about:

- (A) 60000 km
- (B) 300000 km
- (C) 15000 km
- (D) 1500000 km

55. A journey has three legs: 20 km at 40 km/h, 30 km at 60 km/h, and 10 km at 20 km/h. The average speed is:

- (A) 40 km/h
- (B) 42 km/h
- (C) 45 km/h
- (D) 60 km/h

56. To make a pendulum clock **run slow** (lose time), you should:

- (A) shorten the pendulum
- (C) lengthen the pendulum

- (B) use a heavier bob
- (D) use a lighter bob

57. A distance–time graph line becomes steeper as time goes on. The object is:

- (A) slowing down
- (C) moving at constant speed
- (B) at rest
- (D) speeding up

58. A town is 100 km away. A bus leaves at 9:00 travelling 50 km/h; a car leaves at 9:30 travelling 100 km/h, both non-stop. Who reaches the town first?

- (A) the bus
- (C) the car
- (B) both at the same time
- (D) it cannot be decided

59. Which of the following is the **slowest** speed?

- (A) 8 m/s
- (C) 600 m/min
- (B) 36 km/h
- (D) 0.5 km/min

60. Object A moves 100 m in 4 s; object B moves 150 m in 5 s. Which statement is correct?

- (A) A is faster by 5 m/s
- (C) both have equal speed
- (B) B is faster by 5 m/s
- (D) B is faster by 50 m/s

End of question paper. Maximum marks: 240. Marking: +4 correct, –1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 13: Motion and Time

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	D	21	B	31	B	41	D	51	B
2	B	12	B	22	C	32	C	42	C	52	D
3	A	13	D	23	A	33	C	43	B	53	B
4	D	14	A	24	D	34	D	44	A	54	D
5	B	15	B	25	D	35	A	45	A	55	A
6	D	16	C	26	B	36	B	46	B	56	C
7	C	17	C	27	C	37	B	47	C	57	D
8	A	18	A	28	A	38	A	48	D	58	C
9	A	19	D	29	A	39	D	49	C	59	A
10	C	20	B	30	D	40	C	50	A	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Periodic motion repeats at regular intervals; oscillatory motion is the special to-and-fro kind. The Earth's revolution repeats every year (periodic) but is circular, not to-and-fro — periodic and *not* oscillatory. The pendulum, guitar string and swing all oscillate.

2. (B) Leg 1 speed = $100/5 = 20$ m/s; leg 2 = $100/4 = 25$ m/s. Equal distances took unequal times → speed changed → non-uniform. ("Straight line" and "equal distance" do not by themselves mean uniform.)

3. (A) 2 min = 120 s. Speed = $6 \div 120 = 0.05$ m/s. (Dividing by 2 instead of 120 gives the trap 3 m/s.)

4. (D) $72 \times 5/18 = 4 \times 5 = 20$ m/s.

5. (B) $15 \times 3.6 = 54$ km/h. (Dividing by 3.6 gives the reverse-direction trap 4.17.)

6. (D) Equal distances, different speeds → total distance ÷ total time, not the mean of the speeds. Let each leg be d: time = $d/60 + d/40 = 5d/120 = d/24$; average = $2d \div (d/24) = 48$ km/h. (The mean $(60+40)/2 = 50$ is the classic trap.)

- 7. (C)** Anya's time = $60/40 = 1.5$ h. Ravi's time = $30/30 + 30/60 = 1 + 0.5 = 1.5$ h. They tie. (Trap: Ravi's higher top speed makes people pick him, but he loses time on the slow first half.)
- 8. (A)** A slanted line = moving at steady speed; a horizontal line = at rest; a steeper slanted line = moving again but faster. So: moves, then stops, then moves faster.
- 9. (A)** Time period = $40 \text{ s} \div 20 \text{ oscillations} = 2 \text{ s}$.
- 10. (C)** The SI unit of time is the second.
- 11. (D)** $90 \text{ km/h} = 25 \text{ m/s}$; distance in 10 s = $25 \times 10 = 250 \text{ m}$. (900 m forgets the conversion.)
- 12. (B)** Total distance = 50 km; total time = $2 + 1 \text{ (rest)} + 1 = 4 \text{ h}$; average = $50/4 = 12.5 \text{ km/h}$. (Ignoring the rest gives $50/3 \approx 16.7$ — the trap.)
- 13. (D)** A longer simple pendulum swings more slowly $\rightarrow$ its time period increases.
- 14. (A)** A sundial uses the position of the Sun's shadow.
- 15. (B)** Slope = speed; the steeper line (P) has the greater speed, so P is faster.
- 16. (C)** $36 \text{ km/h} = 36 \times 5/18 = 10 \text{ m/s}$. $Y = 12 \text{ m/s} > 10 \text{ m/s}$, so Y is faster.
- 17. (C)** $20 \text{ min} = 1/3 \text{ h}$; distance = $54 \times 1/3 = 18 \text{ km}$. (Forgetting to convert minutes inflates the answer.)
- 18. (A)** Time = $240 \div 80 = 3 \text{ h}$.
- 19. (D)** A sundial needs sunlight to cast a shadow, so it cannot work on a cloudy night. A water clock, sand clock and pendulum clock all work in the dark.
- 20. (B)** A body moving along a circular path shows circular motion.
- 21. (B)** A curve whose slope keeps increasing means speed is increasing $\rightarrow$ non-uniform motion with increasing speed.
- 22. (C)** Times are equal (2 h each), so the mean is valid: distance = $40 \times 2 + 60 \times 2 = 200 \text{ km}$; time = 4 h; average = 50 km/h . (Contrast Q6, where equal *distances* make the mean wrong.)
- 23. (A)** A simple pendulum's period depends on length and gravity, not on the bob's mass. Same length $\rightarrow$ equal periods.
- 24. (D)** Distance rises by 5 m each second (5, 10, 15, 20) $\rightarrow$ speed = 5 m/s , constant $\rightarrow$ uniform motion at 5 m/s .

- 25. (D)** From $t = 2 \text{ s}$ to $t = 4 \text{ s}$ the distance stays $20 \text{ m} \rightarrow$ the object is at rest in that interval.
- 26. (B)** Original period $= 60/30 = 2 \text{ s}$ (not needed for the answer). New period $= 3 \text{ s}$, so oscillations in $60 \text{ s} = 60 \div 3 = 20$.
- 27. (C)** Convert all to m/s : $P = 5$; $Q = 21.6 \div 3.6 = 6$; $R = 300 \text{ m} \div 60 \text{ s} = 5$. $Q (6 \text{ m/s})$ is fastest.
- 28. (A)** Equal distances (180 km each): $t_1 = 180/90 = 2 \text{ h}$, $t_2 = 180/60 = 3 \text{ h}$; total 360 km in $5 \text{ h} \rightarrow 72 \text{ km/h}$. (Mean 75 km/h is the trap.)
- 29. (A)** Convert to km/h : $10 \text{ m/s} = 36 \text{ km/h}$; $30 \text{ km/h} = 30 \text{ km/h}$; $720 \text{ m/min} = 720 \times 60 = 43200 \text{ m/h} = 43.2 \text{ km/h}$. Increasing: $30 \text{ km/h} < 10 \text{ m/s} < 720 \text{ m/min}$.
- 30. (D)** A shorter pendulum has a smaller period $\rightarrow$ more oscillations per minute. Y is shorter, so Y makes more. (Mass is irrelevant.)
- 31. (B)** Speed $= 100 \div 12.5 = 8 \text{ m/s}$.
- 32. (C)** $54 \text{ km/h} = 15 \text{ m/s}$; $2 \text{ min} = 120 \text{ s}$; distance $= 15 \times 120 = 1800 \text{ m}$. ($6480 = 54 \times 120$, mixing units; $900 = 15 \times 60$ forgets a minute.)
- 33. (C)** Distances per second $5, 3, 1$ are decreasing $\rightarrow$ speed is decreasing $\rightarrow$ non-uniform, slowing down (typical of going up a slope).
- 34. (D)** The y -value is the distance from the start; where the two lines meet, both objects are at the same distance from the start at that instant. (Meeting says nothing about equal speeds.)
- 35. (A)** Same distance D : speed_A $= D/2$, speed_B $= D/3$. Since $1/2 > 1/3$, A is faster. Ratio $A : B = (1/2) : (1/3) = 3 : 2$.
- 36. (B)** The odometer records total distance travelled.
- 37. (B)** A flattening curve means the slope (speed) is decreasing $\rightarrow$ the object is slowing down.
- 38. (A)** Total distance $= 120 + 90 = 210 \text{ km}$. Total time $= 2 \text{ h} + 20 \text{ min} + 1 \text{ h } 40 \text{ min} = 2 + 1/3 + 5/3 = 4 \text{ h}$. Average $= 210 \div 4 = 52.5 \text{ km/h}$. (Ignoring the halt and using 3 h gives the trap 70 km/h .)
- 39. (D)** Oscillatory motion is a special periodic motion, so every oscillatory motion is periodic; but a periodic motion can be non-oscillatory (e.g., circular revolution). Hence "every oscillatory motion is periodic, but not every periodic motion is oscillatory."
- 40. (C)** Slope $= (20 - 0) \div (4 - 0) = 5 \text{ m/s}$.

- 41. (D)** Slope = $(30 - 6) \div (8 - 2) = 24 \div 6 = 4$ m/s. (Forgetting to subtract the starting values, e.g. $30/8$ or $6/2$, gives the trap values.)
- 42. (C)** Time = $540 \div 30 = 18$ s. Speed in km/h = $30 \times 3.6 = 108$ km/h.
- 43. (B)** Period = $45 \div 30 = 1.5$ s. 3 min = 180 s $\rightarrow$ oscillations = $180 \div 1.5 = 120$.
- 44. (A)** Total distance = 120 km; total time = 1 + 0.5 (stop) + 1 = 2.5 h; average = $120 \div 2.5 = 48$ km/h.
- 45. (A)** Motion along a straight line is rectilinear motion.
- 46. (B)** Average speed = total distance $\div$ total time. Distance comes from the odometer; total time from a clock. (The speedometer gives only the instantaneous speed.)
- 47. (C)** Distance rises 8 m every 2 s $\rightarrow$ speed = $8/2 = 4$ m/s, constant $\rightarrow$ uniform motion at 4 m/s.
- 48. (D)** Speed_P = $60 \div 1.5 = 40$ km/h; speed_Q = $80 \div 2 = 40$ km/h $\rightarrow$ equal.
- 49. (C)** 90 km/h = $90 \times 5/18 = 25$ m/s. The car (25 m/s) beats the bike (22 m/s) by 3 m/s.
- 50. (A)** Distances rise by 2, 6, 10 m in successive seconds — the gaps grow $\rightarrow$ non-uniform, speeding up.
- 51. (B)** A fan blade goes round and round $\rightarrow$ circular motion. A swinging pendulum goes to-and-fro $\rightarrow$ oscillatory motion. In order: circular and oscillatory.
- 52. (D)** Distance is measured from the start; returning to the time axis (distance 0) means the object has come back to its starting point.
- 53. (B)** 1.5 km = 1500 m; 25 min = 1500 s; speed = $1500 \div 1500 = 1.0$ m/s.
- 54. (D)** Distance = $300000 \times 5 = 1500000$ km.
- 55. (A)** Each leg takes 0.5 h (20/40, 30/60, 10/20). Total distance = 60 km; total time = 1.5 h; average = $60 \div 1.5 = 40$ km/h.
- 56. (C)** A longer pendulum has a longer period, so the clock ticks fewer times each real second and runs slow. To make it run slow, lengthen the pendulum.
- 57. (D)** A line getting steeper means the slope (speed) is increasing $\rightarrow$ speeding up.
- 58. (C)** Bus: $100 \div 50 = 2$ h from 9:00 $\rightarrow$ arrives 11:00. Car: starts 9:30, $100 \div 100 = 1$ h $\rightarrow$ arrives 10:30. The car arrives first. (Trap: the bus's half-hour head start is smaller than the car's one-hour speed advantage.)

59. (A) Convert all to m/s: 8 m/s; 36 km/h = 10 m/s; 600 m/min = 10 m/s; 0.5 km/min = 500 m/min $\approx$ 8.33 m/s. The slowest is 8 m/s.

60. (B) Speed_A = $100 \div 4 = 25$ m/s; speed_B = $150 \div 5 = 30$ m/s. B is faster by $30 - 25 = 5$ m/s.

Solutions complete: 60 / 60.